8.0 Introduction

The most conventional steering arrangement is to turn the front wheels using a hand-operated steering wheel which is positioned in front of the driver to allow it to deviate somewhat from a straight line. The system allows a driver to use only light forces to steer a heavy vehicle. Front axle carries the weight of the front part of the automobile as well as facilitates steering and absorbs shocks due to road surface variations. Let us see about the structure and its functioning in the following segment.

8.1 Functions of the steering system

- Steering system is helps to turn the front wheels on desired direction with the help of steering wheel.
- This system provides to vehicle steer in turning are also stabilize the vehicle on the straight road.

8.2 Parts of steering system

1. Steering Wheel
2. Steering Column
3. Steering Shaft
4. Steering Gear Box
5. Cross Shaft
6. Drop arm (or) Pitman arm
7. Drag Link (or) Pull and Push rod
8. Tie Rod or Track Rod
9. Ball Joint
10. Front Axle
11. Road Wheel

8.2.1 Steering parts and its uses

1. Steering Wheel

It is a circular wheel mounted at the centre portion with two or three rods. The steering wheel is connected with steering gear box through steering shaft. So the road wheels are turns easily when rotation of steering wheel.

2. Steering Column

It is hollow shaft in which is made up on alloy steel. The steering shaft rotates with the help of bearing. In modern vehicles, the gear change lever is provided at the steering column for the convenience of the driver.

3. Steering Shaft

It is made up of good quality steel. The upper end is fixed in steering wheel with the help of splines (or) key. The other end is connected to the steering gear box.

4. Steering Gear Box

The steering gear box serves the following purposes.

1. It provides mechanical advantages and enable the driver to steer the vehicle easily.
2. It converts the turning motion of the steering wheel into the to-and-fro motion of the link rod of the steering linkage.

5. Steering Linkages

- When the steering wheel is turned its motion is carried to the steering gear box through the steering shaft.

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